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Section 1

Question-1

i) Application Layer from the TCP/IP model

ii) Network Access (physical Layer) clock synchronization before transmitting signals over a physical midium is a physical layer funct

iii) Network Access (Data Link) Layer

Question 2

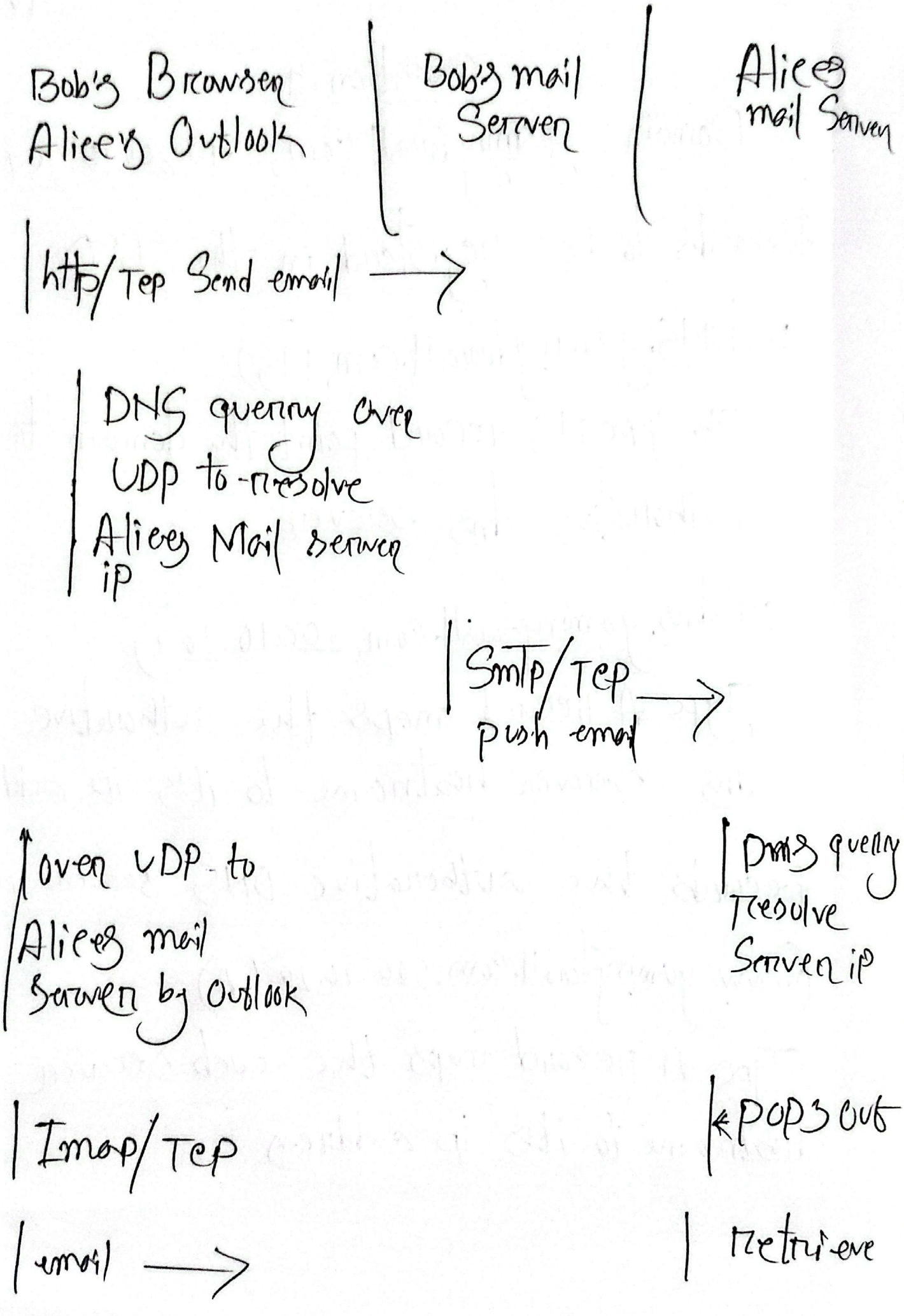
Cookies are browser specific. They are sorted in a cookie file managed by the individual browser on the user's end system.

When the user visits www.news.com using google chrome, the server sent a set-cookie header in the HTTP response & Chrome stored that cookie with a unique id number in Chrome owns cookie file. On subsequent

request chrome include cookie in the Cookie header line
when the user visited the same website later using Internet Explorer IE has its own separate cookie which doesn't Contain the cookie Saved by chrome. Because cookie are not shared between different browser

Question 3

Bob sends via web based email
Alice Received via Outlook
no one knows mail servers ip initially


Bob's Browser
Alice's Outlook
Bob's mail Server
Alice's mail Server
http/Tcp Send email
DNS query over UDP to resolve Alice's Mail server ip
SmTp/Tcp push email
over UDP to Alice's mail Server by Outlook
DNS query resolve Server ip
Imap/Tcp
email
POP3 out
retrieve

Question-4

Domain gamingforall.com, ip 200.10.20.1

Records to be registerd in the TLD

1. (DNS.gamingforall.com, NS)
   ~~The~~ Type NS record point the domain to its authorative dns server

2. (dns.gamingforall.com, 200.10.20.1)
   Type A record maps the authorative dns server hostname to its ip address

Records the authorative DNS server itself

(www.gamingforall.com, 200.10.20.1, A)
Type A record maps the web server hostname to its ip address

Question-5

protocall UDP, Live streaming fifa would cup uses udp beeause its connectionless Low, Latency & Suitable for real time media where occasional data loss is acceptable

port type 60001 is a dynamic/random it fails in range 1024-65535 which is used for client side temporary ports

Since UDP demultiplexing uses only the destination ip & destination port. all segments with the same destination will go into the same Socket. However the request comes from diff Source ip address

Question-6 (1)

for non persistant http each object including the base html file require:

1 TCP RTT (35 ms) + 1 HTTP RTT (30 ms)

= 65 ms per object

total object 35

RTT = 35 (35 + 30) = 35 × 65

= 2275 ms

(11)

file trans time per object = $\frac{\text{Obj Size}}{\text{Server Speed}}$

$= \frac{4\ MB}{64\ MBPS}$

$= \frac{32}{64}$ seconds

= .5 s

= 500 ms / object

total transmission time (base html + 34 object) each 4 MB

= 35 × 500 = 17500 ms

question-7

Lan Delay = 15 ms

30% resolved by BRACU proxy

Dept Lan + BRACU Lan 15+30 = 45 ms

30% resolved by origin server

∴ Dept Lan + BRACU Lan + Access Delay + Internet Delay = 15+30+100+200

345 ms

i) Avg response time

.4×15 + .3×45 + .3×345 = ~~6+135.5~~

= 6 + 13.5 + 103.5

= 123 ms

ii) A device from EEE department just visited xyz.com. That means the the connect is now cached in the

BRACU central proxy server

when peA wants to visit xyz

i) CSE dept proxy → miss

ii) BRACU proxy → hit

Question 8

Client ISN 1910

Server " 1532

$e_1 = 421$ bytes, $e_2 = 320$ bytes, $e_3 = 111$ bytes

$S_1 = 260$ bytes $S_2 = 220$ bytes

client rwnd = 6000 bytes,

Server " = 12000 "

First Handshake

→ Syn client → Server (Seq = 1910)

→ SYN Ack → client (Seq = 1532
ack = 1911)

→ Ack to: client → Server (Seq = 1911
ack = 1533
data = 421 byte)

After Handshake

Isn+1

client data starts at byte 1911

Server " " " " 1533

Segment	Sequence	Ack	Data
C1 (Client-S)	1911	1533	421 byte
S1 (Server-C)	1533	2332	260 byte
S2	1793	2332	220 "
C2	2392		320 "
S1 retransmit	1533		260 "
C3	~~263~~ 2652		111 bytes

~~part~~

(I)

From the server the RTO time for S_1 expires before the server receives an Ack for S_1 The server has not ~~se~~ received ack for S_1 within the Timeout period so, it retransmit S_1.

Since this is selective repeat, the client has already received S_1. when the client receives the duplicate S_1, it will discard the duplicate data and resend the ack for S_1 to the Server

(II)

Sequence number of e_{02}:

e_1 Starts at Seq = 1911

data = 421 bytes

e_2 Seq = 1911 + 421 = 2332

Ack number of e_2

S_1 covers bytes 1533 to 1792 → 1793

S_2 Covers " 1793 to 2012 → 2013

Since both S_1 & S_2 are received in order (S_1 then S_2) & using Selective repeat the client has received all bytes upto 2012

Ack # of e_2 = 1533 + 260 + 220 = 2013

e_2 Sequence = 2332 ack = 2013

(iii)

Server initial rwnd = 12000 bytes

Server has received e_1 (421 bytes) +

e_2 (320 bytes) +

e_3 (111 bytes) = 852 bytes

total from client

rwnd = 12000 − (421 + 320 + 111) = 12000 − 852

= 11148 bytes